



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year 2024-25 (Semester II)

Summer-Term

Date: 21/06/2025

Time: 14.30-17.30 hrs

Subject : PCC-IN206/PCC-IN207 Signals and Systems

Class : S. Y. (Instrumentation Engg.)

Max. Marks : 100

Q. No.		Question	Marks
Q. 1	A	Given the signal $x(t) = 2\cos(10\pi t + \pi/4) + 4\cos(40\pi t + \pi/6)$ Compute the total energy (E_∞) and average power (P_∞) of the signal. Also determine the fundamental period of the signal.	10
	B	Obtain the linear convolution of the following finite length discrete-time signals $x[n]$ and $h[n]$. $x[n] = 2, 1, 2, 3, 3$ $h[n] = 2, 3, 1, 2$ $\uparrow n=0$ $\uparrow n=0$	
Q. 2	A	Let Input-output relationship of an LTI system is $y(t) = tx(t) + 1$ Test the system for the following properties. (i) Linearity (ii) Time-Invariance (iii) Causality (iv) Stability and (v) Memoryless	10
	B	Let Impulse response of an LTI system is $h[n] = (\frac{1}{2})^n u[n]$. Obtain the output of the system to unit step input (i.e. $x[n] = u[n]$)	10
Q. 3	A	Determine the Fourier transform of (i) $x(t) = e^{-3t}u(t)$ (ii) $x[n] = (\frac{1}{4})^n u[n]$	10
	B	Consider a causal LTI system with transfer function $H(s) = \frac{1}{s+3}$ for $\sigma > -3$. Determine the output $y(t)$ of the system to input $x(t) = e^{-2t}u(t)$	10
Q. 4	A	Suppose that a causal LTI system is described by following difference equation $y[n] = \frac{1}{3}y[n-1] + x[n]$. Compute the output of the system for input $x[n] = (\frac{1}{2})^n u[n]$	10
	B	Let a causal LTI system is described by following differential equation. $\frac{dy(t)}{dt} + 2y(t) = x(t)$, Find the response of the system to input $x(t) = e^{-3t}u(t)$.	10
Q. 5	A	The transfer function of a causal LTI system is $H(z) = \frac{z}{(z + \frac{1}{5})}$ for $ z > \frac{1}{5}$. Determine the response of the system to input $x[n] = (\frac{1}{5})^n u[n]$	10
	B	Let the input-output relationship of an LTI system is $y[n] = x[n] + 2x[n-1] + x[n-2] + 3x[n-3]$ Determine the transfer function and impulse response of the system. Also determine the output corresponding to the following input. $x[n] = \{1, 2, 1, 3, 2, 3\}$ $\uparrow n=0$	10

The End

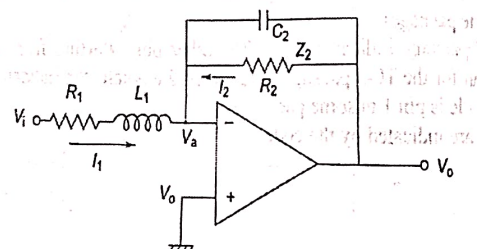


Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Instrumentation Engineering
End-Semester Examination
Academic Year: 2024-25 (Semester – II)

Class: SY BTech.	Subject: Linear Integrated Circuits	Subject Code: PCC-IN204/205
Date: 20.June.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. All questions are compulsory.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Figures to right indicate full marks.

Q.N.	Questions	Marks	CO	BT
Q.1	An op-amp has R_1 and L_1 in series on the input side, connected to the inverting terminal and R_2, connected in parallel in the feedback path. The non-inverting terminal is grounded. Derive the voltage gain 	10	CO1	BT4
Q.2	Design a wide band-pass filter with $f_L = 1\text{kHz}$, and a pass band gain = 4. Draw the frequency response plot for this filter. Calculate the value of the Q for the filter.	10	CO2	BT6
Q.3	Explain wide Band-Reject filter with frequency response.	10	CO1	BT2
Q.4	Derive the frequency of the RC phase shift oscillator in detail.	10	CO2	BT4
Q.5	Explain the 8-bit successive approximation A/D converter with proper example.	10	CO4	BT2
Q.6	Derive the output equation for averaging amplifier and summing amplifier in noninverting configuration.	10	CO1	BT4
Q.7	The 741C op-amp having the following parameters is connected as a non-inverting amplifier with $R_1 = 1\text{k}\Omega$ and $R_F = 10\text{k}\Omega$: $A = 200,000$, $R_i = 2\text{M}\Omega$, $R_o = 75\Omega$, $f_o = 5\text{Hz}$, supply voltages = $\pm 15\text{V}$, output voltage swing = $\pm 13\text{V}$. Compute the values of A_F , R_{iF} , R_{oF} , f_F , and V_{OOT} .	10	CO3	BT3
Q.8	Design a differentiator that will differentiate an input signal with $f_{max} = 100\text{Hz}$.	10	CO2	BT6
Q.9	Derive the output equation for instrumentation amplifier using Transducer Bridge	10	CO2 CO3	BT4
Q.10	Explain the non-inverting comparator with its application and its proper input and output waveform when a) V_{ref} is positive b) V_{ref} is Negative	10	CO1	BT2



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
Summer Term Examination, AY 2024-25
Class: S.Y. B.Tech (Instrumentation Engineering)

Date: 24/06/2025

Subject: **SENSORS AND TRANSDUCERS**
Code: PCC-IN201

Max. Marks: 100

Time: 14.30 to 17.30pm

Sr. No	Question	Marks	COs	BT
Q. 1	a. Define the following terms: 1. Resolution 2. Precision 3. Range 4. Span 5. Repeatability b. Write short note on: 1. Pyrometer 2. Semiconductor strain gauge	(05) (05)	CO1 ,CO3 ,CO3	BT1,BT2 BT1,BT2
Q. 2	a. Describe with a diagram how flow rate of liquids can be measured by venturimeter meter. b. Explain turbine type flowmeter for flow measurement. What type of liquid is suitable for turbine type flowmeter?	(05) (05)	CO4 CO2	BT1 BT1
Q.3	a. Explain the working principle of a capacitive type level measurement with its applications. b. How the level of closed tank is measured? Give the applications.	(05) (05)	CO4 CO2	BT1 BT1
Q. 4	a. What is gauge factor? Give its expression. Explain how strain gauge is giving change in resistance. b. Explain the weighing machine application of strain gauge with its block diagram.	(05) (05)	CO3 CO2 ,CO5	BT1 BT5
Q.5	a. Draw the schematic diagram of a Bellows and explain its operation. b. Explain bourdon tube as a primary sensor and LVDT as a secondary transducer with its diagram and application of it.	(05) (05)	CO3 CO2	BT1 BT1
Q.6	a. Describe the working and construction of Linear Variable Differential Transformer? Draw its neat diagram. b. Compute the capacitance of a parallel plate capacitive transducer. The plate is 500 mm ² and the distance of separations is 200 μm . The transducer is used in air. Permittivity of air is 8.85×10^{-12} F/m.	(05) (05)	CO4 CO1	BT1 BT5
Q.7	a. Describe the working principle and construction of Thermistor with its applications. b. What is seebeck effect? Explain thermocouple with its construction and operation in detail.	(05) (05)	CO2 CO2	BT1 BT5
Q.8	a. Describe the generalized instrumentation system with its block diagram.	(05)	CO1	BT2